

Repricing

Once the initial exchange of repo funds and repo collateral has been made, it's possible, even likely, that the value of the collateral will change before the end of the repo agreement. If the value of the collateral increases, the lender will be holding more collateral than is required to secure the loan and will be asked to return the surplus collateral. If the value of the collateral decreases, the lender will be holding less collateral than is required to secure the loan, and the borrower will be required to post additional collateral. This process is referred to as repricing and occurs throughout the term of the repo transaction along terms set out in the initial repo documentation.⁵

Repricing is an issue too often neglected by analysts when calculating carry and cash flows. For example, imagine a trader has purchased bonds financed via a term repo transaction and has sold bond futures against the long bond position. If the price of the bonds declines, the repo counterparty is likely to ask for more collateral. At the same time, the price of the futures contracts is likely to have decreased, resulting in a positive inflow of cash in the futures margin account. The cash in the futures margin account can be provided directly to the lender to satisfy the call for additional collateral, or the cash could be used to purchase additional bonds, which could be used as collateral. Either way, the additional transaction is likely to have at least some impact on the interest receipts and/or payments made by the trader.

Many larger financial institutions have collateral management desks that consolidate the process of posting collateral, including repricings, insulating the bond trader from the need to worry about these considerations. But that won't be the case for every trader. And even traders who benefit from a central collateral management operation should be aware of the mechanics affecting the financing of their positions.

Specialness

Repo desks typically accept a large number of bonds as collateral for a repo loan, with no real preference for any specific issue. We refer to such bonds as 'general collateral' (GC). However, there are some scenarios where a repo desk or the market in general will have a preference for a particular bond.

For example, consider a scenario in which a trader has borrowed a particular bond from a repo desk so that he could sell the issue short. The repo desk will need to acquire this bond in the market on or before the day it is scheduled

⁵The rules governing acceptable collateral usually are negotiated between counterparties and will specify which instruments are acceptable as collateral and under what conditions repricing is to occur.

to be returned to the original counterparty who previously submitted the bond as collateral.

It may be a simple matter for the repo desk to find that bond and persuade another of its clients to provide that bond as collateral. But that won't always be the case. If the repo desk is finding it difficult to obtain the bond, it can offer to lend funds at a reduced interest rate for any borrower who is able to provide that bond as collateral. In this case, we say the bond has become 'special collateral.' Equivalently, we may say that the bond has 'gone special.'

A bond that confers a borrowing advantage to its owner is worth more than a bond that is trading as GC in the repo market. As a result, for traders to be indifferent between the two bonds, the bond that is special in the repo market needs to trade with a spot price that is greater than that of the GC bond by an amount that exactly offsets the advantage the special bond provides in the repo market.

Some bonds are more likely than others to experience periods of specialness. For example, bonds that are deliverable into futures contracts are more likely to be special than other bonds. Even if a bond isn't part of a futures delivery basket at the moment, the prospect that it will become eligible for delivery and possibly experience a period of specialness may give it additional value in the spot market today. In this sense, repo specialness is asymmetric and therefore exhibits some similarities with call options.

Owning a bond that becomes special is a bit like winning the lottery in this sense. And the price premium that a bond enjoys in the spot market is related to the likelihood and extent of any specialness it might enjoy over its life.⁶

Repo 'Fails'

In the above example, a repo dealer lent a particular bond to a trader who wished to sell the issue short. In this transaction, the trader was lending money, and the repo dealer was lending the bond as collateral to this trader, who was able to sell the bond in the spot market.

The repo dealer in this example was able to provide the bond to the trader because it previously had been posted to the dealer as collateral for another transaction. And at some point, the dealer will need to return this bond to the original counterparty.

Of course, one possibility is that the dealer will receive the bond from the short-seller before he's scheduled to return it to the original counterparty. But that need not be the case. For example, the repo dealer may have entered into a term repo with the short seller for a date that falls after the date on which the collateral is scheduled to be returned to the original counterparty.

⁶For more on repo specialness, see Darrell Duffie (1996), Special Repo Rates. *Journal of Finance*, 51(2), 493–526.

In that case, the repo dealer will have to find the bond elsewhere – perhaps in the portfolio of another customer or perhaps from the street (e.g. through an interdealer broker).

If the repo dealer has trouble finding the bond, he can provide an incentive to the market by lowering the repo rate on the bond. If he's having considerable difficulty, he may have to lower the repo rate quite a bit.

But in some circumstances, despite his best efforts, the dealer simply may not be able to find the bond in time to return the bond to the original counterparty. In that case, the dealer is said to have 'failed' to the original counterparty.

Repo failures typically carry a penalty to provide an incentive against failure, and these penalties have varied over time and in different markets. In some markets, failure could result in a suspension of the repo desk from the market. In some markets, the failure penalty was that the person borrowing money could hold onto the funds at an interest rate of zero until the collateral was returned. Of course, this penalty is more effective when interest rates are high than when they're low.

Haircuts

Since government bond prices can be volatile, repo transactions often require the party borrowing cash to post as collateral bonds having a somewhat greater market value than the cash being lent. The difference between the value of the cash and the value of the bonds is referred to as the haircut of the collateral.

If repo counterparties can fund the haircuts at the repo rate, then the haircuts should be of no consequence when considering valuations along the curve. However, if repo counterparties have to fund the haircuts at a cost of capital in excess of the repo rate, then these haircuts may have some effect on the relative pricing of bonds across the curve.

Haircuts can have a considerable impact on the economics of repo transactions and depend both on changing haircut schedules and on the specific funding situation of individual market participants. Chapter 18 provides more details and examples.

We'll also find that haircuts can have a considerable impact on swap spreads and hence need to be included in models in order to obtain realistic predictions (see Figures 12.4 and 17.2) – which is a difficult task, given that the haircut funding costs of the marginal market participant cannot be observed.

Right of Substitution

In some cases, the party who borrows cash and provides collateral will retain the right to substitute new collateral for existing collateral during the course of the repo loan. In this case, we say that the borrower has the *right of substitution*.

The right of substitution is an option that benefits the cash borrower in the event that the bonds submitted as collateral become special during the term of the repo transaction. Like most options, its value *ex ante* depends on the extent to which the bonds in question are likely to ‘go special.’ As this right is valuable to the party borrowing cash, the owner of this right pays for it by agreeing to pay a higher repo rate to the party lending the cash.

Many analysts ignore the right of substitution in their calculations, and we admit that the issue may seem a bit esoteric in many cases. But there are times when additional relative value can be derived by being smart about this issue, and we suggest being aware of the issue when looking carefully at repo calculations.

Credit: Counterparty and Collateral Combined

One of the key features of repo transactions is that they rely on more than a single source of credit. For example, for a lender of cash in a repo transaction not to get paid requires that the borrower default at the same time that the collateral drops precipitously in value.

Consider the case of a US bank submitting German Bunds as collateral against a loan of EUR cash from a Japanese bank. If the US bank is unable to repay the EUR cash to the Japanese bank at the end of the repo transaction, the Japanese bank should be made whole as a result of the German Bunds it holds as collateral against the loan. The risk to the Japanese bank is that the German Bunds plummet in value at precisely the time the US bank goes bankrupt.

While this scenario is not impossible, it’s typically viewed as being very unlikely, and most repo desks can manage this risk to some extent by judiciously diversifying collateral types with respect to counterparty.

For example, during the euro crisis, many international banks were wary about lending cash to Spanish banks in exchange for Spanish government bonds, given the apparent linkages between the sovereign and financial sectors in Spain. Instead, Spanish banks submitted their Spanish government bonds as collateral against loans of EUR cash from the ECB. If the Spanish banks need cash in other currencies, they’ve been able to use EUR cash as collateral against loans of cash in these other currencies, such as USD, in the form of shorter-term FX swaps or longer-dated cross-currency basis swaps.

Tri-Party or Custodial Repo

In a bilateral repo transaction, the party receiving collateral typically is able to make use of that collateral in a variety of ways. For instance, in the example above, the repo dealer lent bonds that he was holding as collateral to a trader, so that the trader could sell the bond short.

And as we continued with this example above, we saw that the repo dealer might be caught short himself, unable to return the bonds to the original counterparty, resulting in a repo fail.

To continue with this example, let's imagine that this repo dealer went bankrupt before being able to return the collateral to the original counterparty. In this case, the original counterparty may have to spend considerable time, effort, and money in order to replace the missing bonds. Many repo counterparties consider this risk to be small and manageable, but some participants in the repo market are concerned about it.

One way to mitigate this risk is to use a custodial bank to hold the collateral, in which case the custodial bank is a third party to the transaction, which is then referred to as a tri-party repo.

Most relative value analysts tend not to concern themselves with tri-party repos, but there are times when it's useful to consider the implications of tri-party vs. bilateral repo. For example, there have been times when haircuts on bilateral repo transactions have tended to be considerably wider than in the tri-party market. And as we've seen above, the size of any haircut applied in a repo financing typically impacts the all-in cost of carrying a position, as the haircut typically must be financed at a higher cost of funds.

SOFR

Based on the O/N repo market, the Fed calculates and publishes SOFR, the main reference rate in the US. For the details of this process, we refer to Chapter 1 of Huggins and Schaller (2022), and the illustration in Figure 11.2. Note that due to the exclusion of (most) special repo rates during the calculation process, SOFR is usually close to the GC repo rate. Hence, the spread between SOFR and the repo rate of a specific bond is mostly driven by the specialness of that bond and therefore close to zero for the majority of Treasuries.

A major problem of the new reference rate was the tendency for SOFR to exhibit spikes, especially around the end of quarter. On some days, SOFR quoted more than 20 bp above Fed Funds. A secured rate trading above an unsecured rate is an obvious arbitrage opportunity (particularly when executed with the same counterpart). But with banks being subject to balance sheet constraints and regulatory restrictions (see Chapter 18), the threshold for them to execute this arbitrage can be quite high, especially at the end of quarter. Realizing the problem that banks were unable to fulfill their role as arbitrageur and provide liquidity in the repo market precisely at those times when they would be needed most, the Fed decided to take over that function and announced on July 28, 2021, the implementation of a standing repo

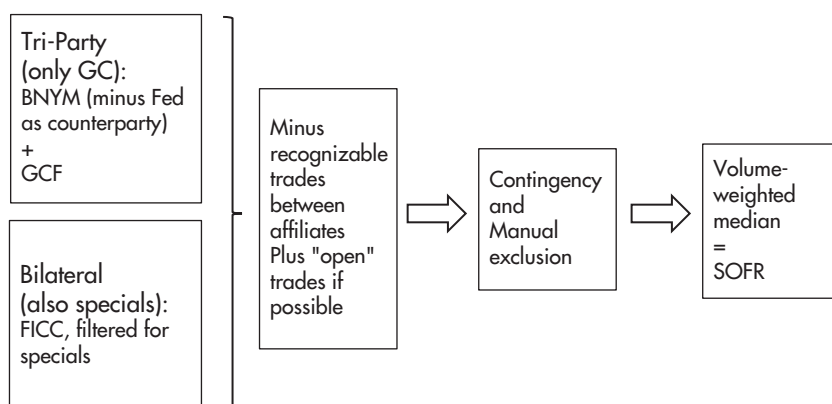


FIGURE 11.2 SOFR calculation.

Source: Huggins and Schaller (2022).

facility (SRF). The SRF provides liquidity to the repo market independent of the balance sheet constraints of banks, with the minimum rate usually being set at the top of the current policy window. In other words, via its repo facility the Fed has introduced the SRF rate (SRFR) as a constraint, i.e. $\text{SOFR} \leq \text{SRFR}$. This had the desired effect and spikes in SOFR have since then disappeared, as shown in Figure 11.3.

However, while the SRF was successful in suppressing spikes, it has introduced a structural break in the SOFR time series before and after the Fed repo facility existed, i.e. without and with the constraint $\text{SOFR} \leq \text{SRFR}$. This causes a problem for analysts requiring a long-term time series of swap rates as basis for their job. In case a ten-year history of USD swap rates is needed, the analyst deals with three different types of reference rates for the swap: LIBOR, SOFR pre-SRF, and SOFR post-SRF. While he may apply ad hoc “solutions” such as adjusting for the average spread between LIBOR and SOFR during the period both traded, we know no method to generate long-term time series for USD swap rates, which can be used as input for analysis requiring a high degree of precision.

Given this situation, we therefore recommend using constant maturity par rates calculated via a fitted curve model as input for RV analysis in those countries, where swaps have experienced structural breaks. For the US in particular, it seems advisable to apply the fitted curve model described in Chapter 8 to generate a time series of par rates for synthetic constant maturity Treasuries, for example, 2Y, 5Y, and 10Y. While this involves some effort, it avoids basing the analysis on swap rate time series with structural breaks.

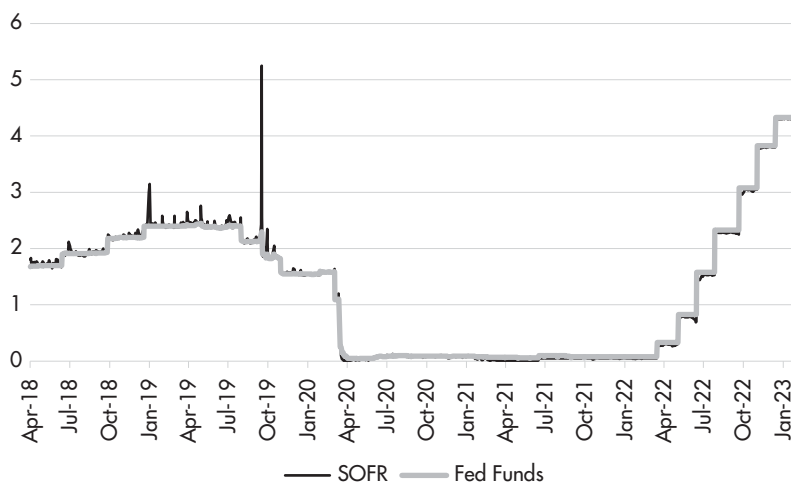


FIGURE 11.3 Evolution of the USD reference rate used by most market participants. *Source:* Authors, from data provided by the Fed on www.newyorkfed.org. Disclaimer: these reference data are subject to the Terms of Use posted at newyorkfed.org. The New York Fed is not responsible for publication of the reference data by the Authors and Wiley and does not endorse any particular publication, and has no liability for your use.

DRIVING FORCES OF THE SPREAD BETWEEN DIFFERENT REFERENCE RATES

As illustrated in Table 11.1, asset and basis swaps involve a number of different reference rates. An understanding of the spread between them is therefore a key element in the analysis of asset and basis swaps provided in the following chapters. According to the classification of reference rates by the two criteria, secured or unsecured and term or O/N, this task can be broken down into two components:

- Understanding the driving forces between unsecured O/N and secured O/N reference rates, i.e. the second and third column of Table 11.1. The different BIS risk weighting results in a significant benefit of secured lending. While secured lending also improves the credit exposure, the difference is usually negligible for O/N lending to contributing banks. Hence, the main advantage of secured lending emerges *indirectly* via capital requirements rather than *directly* via less default risk. Correspondingly, the spread between secured and unsecured O/N rates is mainly a

function of the cost of capital: the higher the cost of capital, the higher the preference for secured lending with 0% BIS risk weighting.

- Understanding the driving forces between unsecured term and unsecured O/N rates, i.e. the first and second column of Table 11.1. As both have the same BIS risk weighting, the cost of capital has no impact on this spread. On the other hand, the longer the term, the higher the credit risk: whereas lending to a contributing bank on an O/N basis may involve negligible credit exposure, the default risk may become considerable for a 6M term.

With some simplifications, one can therefore say that the spread between secured and unsecured O/N rates is driven by the cost of capital, while the spread between unsecured O/N and term rates is driven by credit. Using this conceptual approach, we will now fill in the details.

THE SPREAD BETWEEN SECURED AND UNSECURED LOANS AS DRIVEN BY CAPITAL REQUIREMENTS

In most regulatory jurisdictions, the capital requirements that banks face for unsecured loans are different from the requirements they face for secured loans. These requirements typically conform to the Basel directives, which have evolved over time since the Basel capital accords of 1998.

In general, unsecured lending between banks carries a risk weighting under Basel rules. The precise risk weights depend on a variety of factors but usually involve a minimum risk weighting of 20% in most cases. So, for example, an asset with a 20% risk weight would require capital of 1.6% for a bank with a core capital requirement of 8%. In this case, 1.6% of the interbank loan needs to be funded at the lending bank's cost of equity (or other core) capital. The Bank for International Settlements' (BIS) risk weighting for a secured repo transaction, however, is less in most cases. In fact, for repos involving high-quality government bonds as collateral, the BIS risk weighting is zero in many jurisdictions.

Independent of the actual credit risk – which may well be negligible for lending O/N to high-quality banks – these capital requirements introduce a significant benefit for lending on a secured basis. Assuming a bank has a cost of equity of 10%, lending unsecured results in $10\% * 1.6\% = 16$ bp capital costs. Hence, this bank would prefer (speaking from a funding perspective only) lending secured until the repo rates are more than 16 bp below the rates for unsecured lending. If this bank is the marginal market participant, the spread between unsecured (e.g. Fed Funds or ESTR) and (GC) repo rates should therefore be 16 bp (again, seen only from a funding perspective).

As a starting point, one can therefore estimate the spread between unsecured and secured O/N rates by $g*q*d$, where g is the cost of equity capital, q is the BIS risk-weighting for interbank loans, and d is the BIS guideline for the capital adequacy ratio for a 100% risk-weighted asset (typically 8%).

Since both the cost of equity of individual banks and the marginal market participant are impossible to observe *ex ante*, this formula can seldom be used for exact quantifications. However, it explains the qualitative relationships observed: in times of stress in the banking sector, the cost of bank capital increases, hence the bank's preference for repo lending increases, hence the asset swap spread of government bonds versus LIBOR or euro short-term rate (ESTR) decreases.⁷

While changes in g are difficult to quantify, changes in the other variables, q and d , are obvious. In case of discussions about reforming the BIS rules, this formula therefore allows exact predictions. As a hypothetical example, if an increase of the capital adequacy ratio for 100% risk weighted assets (i.e. d) from 8% to 12% was proposed, one could easily forecast Bund asset swap spreads versus LIBOR or ESTR to decrease by 50%, for example, from -20 bp to -30 bp. Note that, in contrast, asset swap spreads versus SOFR are not affected by changes in any of the three variables, g , q , and d .

We will expand this approach to a model for the repo–LIBOR spread by adding a credit component, which we discuss next. Moreover, following the financial crisis, additional capital requirements have been introduced, which vary by jurisdiction and market participant, as discussed in Chapter 18. The general approach developed here therefore potentially requires modifications to reflect the specific regulatory framework.

THE SPREAD BETWEEN UNSECURED O/N AND TERM RATES AS DRIVEN BY CREDIT EXPOSURE

The Effect of Short Rate Expectations

The most obvious effect of a longer tenor is that expectations of future short rates are reflected in LIBOR, whereas expected future short rates are not incorporated into O/N rates.⁸ Over long periods of time, short rates appear to

⁷We use the convention Reference Rate + X for the asset swap spread X. For example, when a Bund quotes at EURIBOR -10 bp, the asset swap is -10 bp (not $+10$ bp). Under this convention, a richening of bonds versus swaps results in lower asset swap spreads.

⁸Strictly speaking, the expectation of high short rates in the remainder of a current reserve maintenance period could induce banks to pay a higher rate on overnight funds, but since this effect is symmetric and not systematic it will not be addressed in this analysis.

increase as often as they decrease, so this effect should not explain persistent differences between O/N rates and LIBORs over long periods of time or persistent differences between longer-dated LIBOR and OIS swap rates.

The Effect of Interest Rate Compounding

Another difference is that O/N rates must be compounded in order to be compared to LIBORs of longer tenors. For example, even if O/N rates were certain to be 5% every day for an entire year, the six-month (6M) LIBOR rate would be 4–5 basis points (bp) above this rate to reflect the difference in the compounding frequencies.

The Effect of Interest Rate Compounding Convexity

One additional conceptual difference relates to the fact that compounding of any frequency introduces a convexity that links interest rate volatility to the expected growth rate in overnight deposits. For example, consider two scenarios in which the expected O/N rate was 5% indefinitely: one in which the volatility around this rate was high, and another in which the volatility of the O/N rate was low. In each scenario, an investor makes the same initial deposit, which is re-invested every night at the prevailing O/N rate in each scenario for an investment horizon of six months.

The expected terminal value of this deposit is greater in the high-volatility scenario than in the low-volatility scenario, simply due to the convexity inherent in the compounding process. All things being equal, longer-term rates should be lower in high-volatility environments than in low-volatility environments. And in the presence of any volatility at all, there is a conceptual downward bias in longer-term rates for this reason. In theory, this convexity effect might lower LIBORs relative to OIS rates. However, the magnitude of this effect for the tenors considered here is very small (much less than a basis point), and this effect will not be addressed further in this book.

The Effects of Risk and Term Premia

Of course, risk premia and term premia are often invoked to explain the difference between overnight rates and rates with longer tenors. These terms are often used by market analysts as catchall terms to refer to the additional return or yield that investors might demand to be compensated for various risks.

In this book, these terms are reserved for more academic usage. In particular, these premia refer here to the compensation that investors demand in the expected return for accepting systematic risks (e.g. consumption risk) resulting from the covariance between the marginal utility of the investor and the return of the asset.